

Study Alignment Strategy

Mapping Lunar-Clean Progress to the Extended Thermal Mapping Study Plan

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Executive Summary. Your 1D finite-difference solver (Lunar-Clean) with Apollo 15/17 validation, 17 publication figures, DEM-based horizon ray-tracing, and a near-complete Icarus manuscript maps directly onto **Phase 1** of the study plan and partially onto **Phase 2**. The key strategic moves are: (1) upgrade from the discrete 3-layer model to the continuous H-parameter + low-T conductivity correction, (2) scale up your existing ray-tracing from 80 m to 5-20 m DEMs with view-factor scattering, and (3) run the spatial thermal map pipeline to produce subsurface temperature products your TSUKIMI colleagues need for their radiative transfer model. Estimated time to a publishable spatial result: 4-5 months from now.

1. What You Already Have

Before planning next steps, here is an honest inventory of your existing assets and how each maps to the study plan phases.

Asset	Status	Maps to Phase
1D finite-difference thermal solver (Crank-Nicolson, Numba JIT)	Done	Phase 1
Discrete 3-layer regolith (0-2 cm / 2-20 cm / >20 cm)	Needs upgrade	Phase 1
Apollo 15 + 17 HFE validation (RMSE ~3-6 K)	Done	Phase 1
17 publication-quality figures (Jupyter notebook)	Done	Phase 5
Horizon ray-tracing on 80 m LOLA DEMs (72 azimuths, Numba)	Done	Phase 2 (partial)
Sky view factor computation	Done	Phase 2 (partial)
Solar geometry (zenith, azimuth, hour angle)	Done	Phase 2
Icarus manuscript (elsarticle LaTeX, single-column review)	Draft ~80%	Phase 5
Lunar-Clean Python package structure	Done	Phase 5
View-factor / secondary illumination (scattered + thermal IR)	Not started	Phase 2
Continuous H-parameter density profile	Not started	Phase 1
Low-T conductivity correction (Martinez-Siegler 2021)	Not started	Phase 1
Spatial thermal map pipeline (pixel-by-pixel 1D runs)	Not started	Phase 3
Ice stability depth mapping	Not started	Phase 4

Your completion rate on Phase 1 is roughly **60%** (solver + Apollo validation done, but property model needs upgrading). Phase 2 is roughly **40%** (ray-tracing done, but no view-factor scattering or high-res DEMs). Phases 3-4 are 0%.

2. Strategic Alignment: What to Do Next

The study plan has 5 phases over ~10 months. Given your September graduation timeline and existing progress, here is a compressed, realistic schedule that leverages what you already have.

Phase 1: Foundation Strengthening (Now - May 2026)

What the plan says: Implement Hayne H-parameter model, Martinez-Siegler low-T correction, Burger microphysical model. Validate against Apollo + Chang'E-4 + ChaSTE.

What you already have: Your discrete 3-layer model works but uses fixed density jumps at 2 cm and 20 cm. Your solver infrastructure (Crank-Nicolson, Numba, boundary conditions) is solid.

What to do now (2-3 weeks):

- Replace the 3-layer density with the continuous H-parameter: $\rho(z) = \rho_d - (\rho_d - \rho_s)\exp(-z/H)$. This is ~20 lines of code in your property functions.
- Add the Martinez-Siegler (2021) low-T conductivity correction as a toggle. This modifies your k_c term below 150 K.
- Keep your existing discrete model as a comparison case — the Icarus paper can show both and quantify the difference.
- Add Chang'E-4 (Von Karman, 471 kg/m³ surface density) and Chandrayaan-3 ChaSTE (69.4°S, K = 0.0115 W/m/K at 80 mm) as additional validation points.
- Priority: this is the single highest-value upgrade for your paper.

Phase 2: Illumination and Shadow Module (May - June 2026)

What the plan says: Build DEM-based illumination with horizon lookup, secondary illumination via view factors, validate against Diviner polar maps.

What you already have: Your Numba-accelerated horizon ray-tracing on 80 m DEMs with 72 azimuthal directions and SVF computation. Solar geometry module.

What to do now (3-4 weeks):

- Download 5-20 m south pole LOLA DEMs from PGDA (pgda.gsfc.nasa.gov/products/78). Your existing ray-tracing code should work with minor I/O changes.
- Implement simplified view-factor secondary illumination. Start with the nearest-neighbor approach from Schorghofer's Topo3D module (open source). This adds reflected sunlight + thermal IR from surrounding terrain into PSRs.
- Use SPICE/HORIZONS ephemeris instead of analytical solar geometry for proper obliquity/precession treatment over the 18.6-year cycle.
- Validate by comparing your illumination fraction map against Mazarico et al. (2011) published products.
- This is where your TSUKIMI connection matters — your colleagues' radiative transfer model needs the thermal profile output from this pipeline.

Phase 3: Spatial Thermal Mapping (June - July 2026)

What the plan says: Couple illumination module with 1D solver pixel-by-pixel. Produce surface + subsurface temperature maps for Artemis sites. Validate against Diviner.

What you already have: Nothing here yet, but your solver + ray-tracer are the two inputs to this pipeline.

What to do (4 weeks):

- Build the driver script: for each DEM pixel, compute local illumination history → run 1D solver → store temperature profiles at key depths (surface, 2 cm, 10 cm, 50 cm, 1 m).
- Use the Fourier-matrix method from Hayne's heat1d for fast steady-state (~100 ms/pixel) rather than spinning up each pixel individually. Alternatively, your Numba solver with 10 spin-up cycles should handle a 500×500 pixel patch in a few hours.
- Start with a small test region (~5×5 km) near Shackleton or de Gerlache. Generate $T_{\max}$, $T_{\min}$, T_{mean} maps.
- Compare synthetic Diviner observations (convolve your model to Diviner footprint and spectral response) with actual Diviner bolometric maps.
- This is the core deliverable for TSUKIMI: spatially-resolved subsurface temperatures as input to the terahertz RTM.

Phase 4: Ice Stability Mapping (July - August 2026)

What the plan says: Compute sublimation rates, map ice stability zones, investigate sensitivity to conductivity model choice.

What you already have: Your notebook already has fig17_ice_stability.png — a preliminary polar ice stability map. But it uses the discrete model, not spatially-resolved profiles.

What to do (3 weeks):

- From your Phase 3 temperature maps, compute time-averaged sublimation rates at each depth using Schorghofer-Williams (2020) methodology.
- Map ice stability depth (shallowest depth where sublimation timescale > 1 Gyr) at each pixel.
- Key novel contribution: run the same mapping with H-parameter, Martinez-Siegler, and Burger property models. Quantify how ice stability depth shifts (5-10 K subsurface temperature differences → potentially km²-scale differences in stable area).
- This sensitivity analysis is the paper's strongest unique contribution that hasn't been done in the literature.

Phase 5: Writing and Dissemination (August - September 2026)

What you already have: An 80%-complete Icarus manuscript, 17 figures, and the Lunar-Clean package on Gitea.

What to do (4 weeks):

- Restructure the Icarus manuscript to include the spatial mapping results and the conductivity model intercomparison.
- The current manuscript focuses on the discrete 3-layer model at Apollo sites. Pivot to: 'From 1D validation to spatial thermal maps: implications for ice stability at Artemis south pole sites.'
- This reframing transforms it from a methods paper into a results paper with direct mission relevance — much stronger for Icarus or PSJ.
- Package Lunar-Clean on GitHub (public) with Zenodo DOI.
- If targeting a PhD in Germany with Prof. Reiß at TUM, this package + paper is your strongest application material.

3. Compressed Timeline (April - September 2026)

Apr 14 - May 4	H-parameter + Martinez-Siegler low-T conductivity implementation	Updated solver with switchable property models	Critical
May 5 - May 18	Add Chang'E-4 + ChaSTE validation; compare discrete vs continuous	3 new validation figures; updated manuscript Section 3	Critical
May 19 - Jun 8	Download 5-20 m DEMs; upgrade ray-tracer; implement view-factor scattering	Illumination module with secondary radiation	High
Jun 9 - Jun 22	Validate illumination against Diviner + Mazarico published maps	Illumination validation figure	High
Jun 23 - Jul 20	Build spatial mapping pipeline; run Shackleton/de Gerlache test region	Surface + subsurface T maps at 4 depths	Critical
Jul 21 - Aug 10	Ice stability mapping + conductivity model sensitivity analysis	Ice stability depth maps; sensitivity comparison	High
Aug 11 - Sep 1	Manuscript rewrite; reframe as spatial results paper	Revised Icarus/PSJ manuscript	Critical
Sep 1 - Sep 15	Code packaging; Zenodo DOI; thesis finalization	Public GitHub repo; thesis submission	Standard

4. Three Decisions to Make Now

Decision 1: Paper scope pivot.

Your current Icarus manuscript is framed around the discrete 3-layer model validated against Apollo. This is solid but incremental. If you add the spatial mapping + conductivity sensitivity results, the paper becomes significantly stronger: 'We show that the choice of thermal conductivity model shifts ice stability depths by X meters and stable area by Y km² at Artemis candidate sites.' Recommendation: pivot the paper scope now rather than after thesis submission.

Decision 2: Target journal.

With the spatial mapping results, you have two viable targets. Icarus (your current target) is strong for methods + validation papers. The Planetary Science Journal (PSJ) is open-access, fast turnaround (~3 months), and increasingly the preferred venue for Artemis-relevant thermal modeling (Schorghofer, Hayne, and Hayes all publish there). Recommendation: PSJ is the stronger fit for a results-focused paper with mission relevance.

Decision 3: TSUKIMI integration depth.

Your TSUKIMI colleagues need subsurface thermal profiles as input to their terahertz radiative transfer model. You can either (a) deliver temperature grids as standalone data products (NetCDF/HDF5) and

let them handle the coupling, or (b) build a joint pipeline where your thermal solver feeds directly into their RTM code. Option (a) is faster and gives you a clean, citable deliverable. Option (b) is more impactful but risks scope creep. Recommendation: start with (a), offer (b) as future work.

5. Specific Code Changes Required

Here are the exact modifications to your Lunar-Clean codebase, ordered by priority.

5.1 Replace density function (Phase 1, ~20 lines)

In your solver module, replace the 3-layer density with:

```
def density_hayne(z, rho_s=1100, rho_d=1800, H=0.06):  
    return rho_d - (rho_d - rho_s) * np.exp(-z / H)
```

Keep your discrete model as `density_discrete()` for comparison. $H = 0.06$ m is the global mean; you can later fit per-pixel from Diviner.

5.2 Add low-T conductivity (Phase 1, ~15 lines)

Martinez-Siegler (2021) modify K_c below ~150 K:

```
def k_solid_martinez(T, rho, chi=2.7):  
    K_c_base = kc_from_density(rho) # your existing function  
    # Reduce solid conduction at low T  
    f_T = np.where(T < 150, (T / 150)**1.5, 1.0)  
    return K_c_base * f_T + chi * (rho / 1000) * T**3
```

The exact functional form is in their paper (Eq. 7-9). The $T^{1.5}$ shown here is an approximation; read the paper for the precise fit.

5.3 View-factor scattering (Phase 2, ~100 lines)

For each pixel i receiving scattered radiation from surrounding pixels j :

$$Q_{\text{scat},i} = \sum_j F_{ij} [\alpha_j S_j \cos(\theta_j) + \epsilon_j \sigma T_j^4]$$

where F_{ij} is the view factor. For a first implementation, approximate F_{ij} analytically using the flat-facet formula and limit the summation to neighbors within ~500 m. Schorghofer's Topo3D code (Fortran, open source) provides a reference implementation you can port to Python.

5.4 Spatial pipeline driver (Phase 3, ~50 lines)

The driver loops over DEM pixels, calls the illumination module and 1D solver:

```
for i, j in pixel_iterator(dem):  
    illum = compute_illumination(i, j, horizons, ephemeris)  
    Q_scatter = compute_scattering(i, j, view_factors, T_surface)  
    T_profile = run_1d_solver(illum, Q_scatter, properties)  
    results[i, j, :] = T_profile[output_depths]
```

Parallelize with `joblib` or `multiprocessing`. A 500×500 pixel patch at 20 m resolution covers a 10×10 km region.

6. PhD Application Strategy Connection

If you apply to TUM with Prof. Reiß, the spatial thermal mapping work directly aligns with their planetary exploration instrumentation focus. Here is how to position each deliverable:

Deliverable	How it helps your PhD application
Published Icarus/PSJ paper with Lunar-Clean	Demonstrates independent research capability and publication track record
Open-source Lunar-Clean package on GitHub	Shows software engineering skills; Prof. Reiß can evaluate code quality directly
Spatial thermal maps for Artemis sites	Directly relevant to TUM's involvement in ESA lunar programs (PROSPECT, etc.)
Conductivity model sensitivity analysis	Shows scientific rigor and ability to identify/quantify model uncertainty
TSUKIMI RTM integration	Demonstrates collaborative, multi-institution research experience

7. Your Next 72 Hours

To build momentum immediately:

1. Download the Martinez-Siegler (2021) paper and their MATLAB code from Zenodo (zenodo.org/records/12586656). Read Sections 2-3 to understand their low-temperature conductivity modification.
 2. In your Lunar-Clean solver, create a new function `conductivity_martinez()` alongside your existing one. Run both at latitude 80°S and compare subsurface temperatures at 1 m depth. You should see a 5-10 K difference.
 3. Download Schorghofer's Planetary-Code-Collection from GitHub. Read the Topo3D/README to understand his view-factor implementation. This is the code you will port to Python for Phase 2.
 4. Download the 20 m south pole LOLA DEM from PGDA. Load it in Python and run your existing horizon ray-tracer on a small 100×100 pixel patch to verify compatibility.
 5. Email Prof. Kasai and Prof. Kamada with a one-paragraph update: 'I am extending Lunar-Clean from 1D Apollo validation to spatial thermal mapping at south pole sites, which will produce the subsurface temperature inputs our TSUKIMI colleagues need for their radiative transfer model.'